



SQL Server

Sources

Massachusetts General Hospital Data Warehouse Diagram

Data Warehouse

Consume



CSV Files

- 1- encounters
- 2- patients
- 3- payers
- 4- procedures

Bronze layer



Raw Data

Object Type: Tables
Load:

- Batch Processing
- Full Load
- Truncate & Insert

Transformations:

- None (as-is)

Data Model: None

Silver layer



Cleansed Data

Object Type: Tables
Load:

- Batch Processing
- SCD Type 1 upsert (MERGE) – patients, payers
- Incremental Insert – encounters
- Truncate & Insert – procedures

Transformations:

- Trim & NULL handling / placeholders
- Standardization (Title Case, gender / marital mapping)
- Deduplication (ROW_NUMBER)
- Referential-integrity checks
- Surrogate keys (IDENTITY / lookups)
- IQR outlier capping (durations)
- Derived columns (duration, patient_name)

Data Model: None

Gold layer



Business-Ready Data

Object Type: Views & Tables
Load:

- Batch Processing
- Full Load

Transformations:

- Data Integration
- Business Logic

Data Model: Star Schema (dims + facts)



Power BI Dashboards



SQL Ad-Hoc Queries



Jupyter Notebooks
(advanced statistical analysis)